

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME: COMPUTER NETWORKS**  
**COURSE CODE: CSA0703**

---

### **COURSE OUTCOME COVERED**

**CO5:** Measure network performance using key metrics with simulation tools.  
(BL4,BL5)

*Assessment Tool Weightage: 30%*

---

### Annexure A

## **Pseudocode Writing Task (Algorithm Design)**

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based

on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

**RUBRICS FOR CALCULATION**

| Criteria                  | Weightage | Level 4 – Exemplary                                                   | Level 3 – Proficient                      | Level 2 – Developing         | Level 1 – Beginning             |
|---------------------------|-----------|-----------------------------------------------------------------------|-------------------------------------------|------------------------------|---------------------------------|
| Problem Understanding     | 20%       | Identifies all inputs, outputs, constraints accurately                | Minor missing elements                    | Partial understanding        | Incorrect interpretation        |
| Algorithm Design          | 30%       | Complete, correct, and logically sound solution                       | Minor logical gaps                        | Partially correct logic      | Incorrect approach              |
| Use of Control Structures | 20%       | Efficient and appropriate use of loops, conditions, functions/modules | Minor inefficiencies                      | Limited or basic usage       | Incorrect or missing constructs |
| Pseudocode Structure      | 10%       | Well-structured, clear, consistent format, easy to follow             | Mostly clear with minor issues            | Difficult to follow in parts | Poorly structured and unclear   |
| Completeness              | 20%       | Handles all cases; optimized approach                                 | Handles most cases; reasonable efficiency | Handles basic cases only     | Incomplete or inefficient       |

## Answers:

### 1. Pseudocode to Calculate Network Throughput and Packet Loss

#### Algorithm

```
START
INPUT totalBits = 8000
INPUT time = 4
INPUT packetsSent = 200
INPUT packetsReceived = 180
IF time == 0 OR packetsSent == 0 THEN
    PRINT "Error: Time and packets sent must be greater than zero."
    STOP
END IF
throughput = totalBits / time
packetLoss = ((packetsSent - packetsReceived) / packetsSent) * 100
PRINT "Throughput =", throughput, "bps"
PRINT "Packet Loss =", packetLoss, "%"
IF throughput > 1500 AND packetLoss < 10 THEN
    status = "Efficient"
ELSE
    status = "Inefficient"
END IF
PRINT "Network Status =", status
PRINT "----- Network Performance Summary -----"
PRINT "Total Bits Received :", totalBits
PRINT "Time :", time, "seconds"
PRINT "Packets Sent :", packetsSent
PRINT "Packets Received :", packetsReceived
PRINT "Throughput :", throughput, "bps"
PRINT "Packet Loss :", packetLoss, "%"
PRINT "Overall Network :", status
STOP
```

#### Output

- Throughput = **2000 bps**
- Packet Loss = **10%**
- Network Status = **Inefficient** (packet loss is **not less than 10%**)

### 2. Pseudocode for Bandwidth Utilization Monitoring

#### Algorithm

```
START
INPUT numberOfLinks
CREATE array bandwidth[numberOfLinks]
```

```

WHILE TRUE
  FOR i = 1 TO numberOfLinks
    INPUT bandwidth[i]
    PRINT "Link", i, "Bandwidth Utilization =", bandwidth[i], "%"
    IF bandwidth[i] > 80 THEN
      PRINT "Warning: Link", i, "is overloaded."
      PRINT "Recommendation: Switch traffic to backup link."
    ELSE
      PRINT "Link", i, "is operating normally."
    END IF
  END FOR
  WAIT for monitoring interval
END WHILE
STOP

```

#### **Explanation**

- Stores bandwidth utilization of all links in an array.
- Uses a loop for continuous monitoring.
- Detects links with utilization above **80%**.
- Recommends switching traffic to a backup link.
- Helps balance traffic, reduce congestion, and improve network performance.

### **3. Pseudocode for Best Path Selection and Fault Tolerance**

#### **Algorithm**

```

START
INPUT numberOfLinks = 6
CREATE arrays
bandwidth[6]
delay[6]
utilization[6]
status[6]
qualityScore[6]
WHILE TRUE
  FOR i = 1 TO 6
    INPUT bandwidth[i]
    INPUT delay[i]
    INPUT utilization[i]
    INPUT status[i]
    qualityScore[i] =
      bandwidth[i] - delay[i] - utilization[i]
  END FOR
  bestPath = 1
  FOR i = 2 TO 6

```

```
    IF qualityScore[i] > qualityScore[bestPath] THEN
        bestPath = i
    END IF
END FOR
PRINT "Best Path =", bestPath
FOR i = 1 TO 6
    IF utilization[i] > 80 OR status[i] == "Failed" THEN
        PRINT "Alert: Link", i, "is overloaded or failed."
        PRINT "Selecting alternate path."
    END IF
END FOR
WAIT for monitoring interval
END WHILE
STOP
```

### Explanation

- Stores bandwidth, delay, utilization, and status for each link.
- Continuously monitors all links.
- Calculates a simple **link quality score** using:
- $\text{Quality Score} = \text{Bandwidth} - \text{Delay} - \text{Utilization}$
- Selects the path with the highest score.
- Detects overloaded (>80% utilization) or failed links.
- Automatically chooses an alternate path and generates an alert.

### Advantages

- Improves **network reliability** by avoiding failed links.
- Enhances **routing efficiency** by selecting the highest-quality path.
- Provides **fault tolerance** through automatic path switching.
- Adapts to changing network conditions with continuous monitoring.